Customer Retention & Churn Analysis

1. Introduction

Customer churn is a major challenge for subscription-based businesses, as it directly impacts revenue and growth. This project focuses on analyzing customer data to identify churn patterns, key retention drivers, and customer lifetime trends.

The objective of this analysis is to understand why customers leave the service and provide actionable recommendations to improve customer retention.

---

2. Dataset Overview

The dataset used in this project contains **30 customer records** representing a subscription-based business model.

Key Features:

- CustomerID – Unique customer identifier
- Gender – Male/Female
- Tenure – Duration of customer relationship (in months)
- MonthlyCharges – Monthly subscription cost
- TotalCharges – Total amount spent
- Contract – Subscription type (Month-to-month, One year, Two year)
- PaymentMethod – Mode of payment
- InternetService – Type of service (DSL / Fiber optic)
- Churn – Whether the customer left (Yes/No)

---

3. Key Metrics

- **Total Customers:** 30
- **Churn Rate:** 40%
- **Retention Rate:** 60%
- **Average Customer Lifetime:** 11.97 months

These metrics indicate a relatively high churn rate, suggesting the need for improved retention strategies.

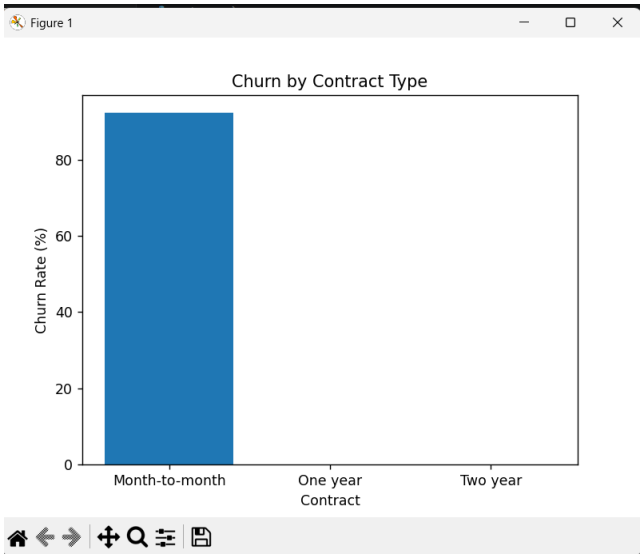
---

4. Analysis

🔹 Churn by Contract Type

- Month-to-month contracts show **92.3% churn**
- One-year and two-year contracts show **0% churn**

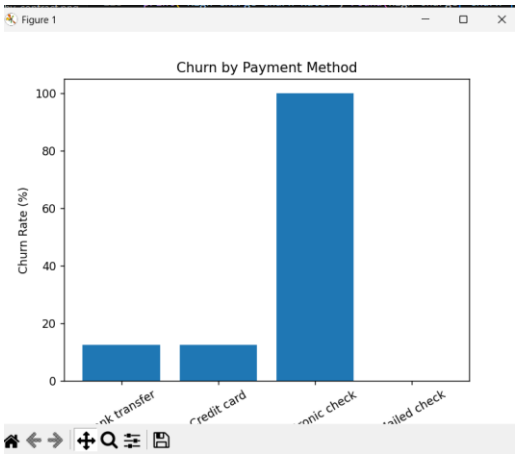
👉 Customers with short-term contracts are significantly more likely to leave.



🔹 Churn by Payment Method

- Electronic check → **100% churn**
- Credit card / Bank transfer → **12.5% churn**
- Mailed check → **0% churn**

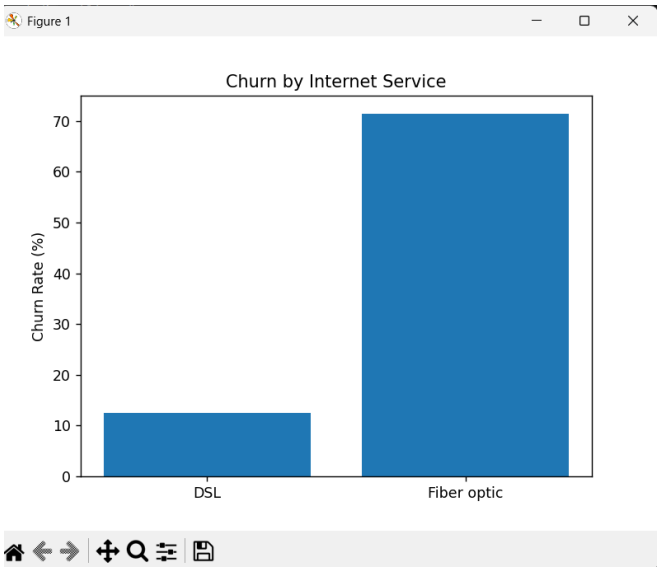
👉 Payment method plays a crucial role in customer retention.



🔹 Churn by Internet Service

- Fiber optic → **71.4% churn**
- DSL → **12.5% churn**

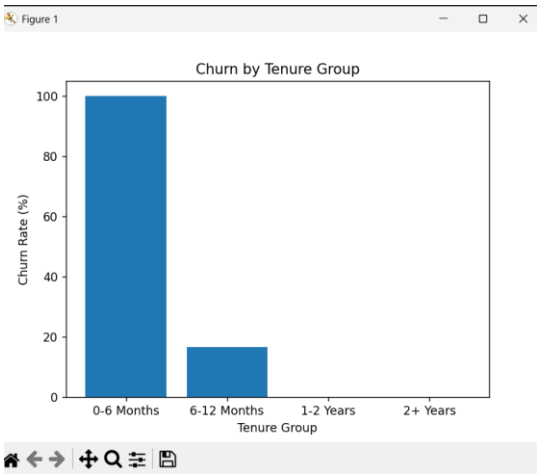
👉 Customers using fiber optic services show higher churn rates.



🔹 Churn by Tenure (Cohort Analysis)

- 0–6 months → **100% churn**
- 6–12 months → **16.7% churn**
- Above 1 year → **0% churn**

👉 Early-stage customers are at the highest risk of churn.



🔹 **Pricing Impact**

- High monthly charges → **66.7% churn**
- Low monthly charges → **13.3% churn**

👉 Higher pricing is associated with increased churn.

🔹 **Revenue Impact**

- Estimated monthly revenue loss due to churn: **947**

👉 Customer churn directly impacts business profitability.

---

5. Key Insights

- Customers on **month-to-month contracts** are highly likely to churn
- **New customers (0–6 months)** are the most vulnerable
- **Electronic check users** exhibit the highest churn behavior
- **Fiber optic users** churn more compared to DSL users
- **Higher pricing plans** lead to increased customer loss

---

6. Recommendations

1. Improve Early Customer Experience

Focus on the first 6 months onboarding and provide tutorials, support, and engagement emails.

2. Promote Long-Term Contracts

Offer discounts for yearly plans and introduce loyalty rewards.

3. Fix Payment Issues

Encourage customers to use credit card or auto-pay methods and reduce dependency on electronic checks.

4. Optimize Pricing Strategy

Re-evaluate high-cost plans and introduce flexible pricing options.

5. Improve Fiber Service Quality

Investigate performance issues and address customer complaints.

---

7. Conclusion

This analysis highlights critical factors affecting customer churn and provides actionable insights to improve retention. By focusing on early customer engagement, pricing strategies, and service quality, businesses can significantly reduce churn and enhance customer lifetime value.